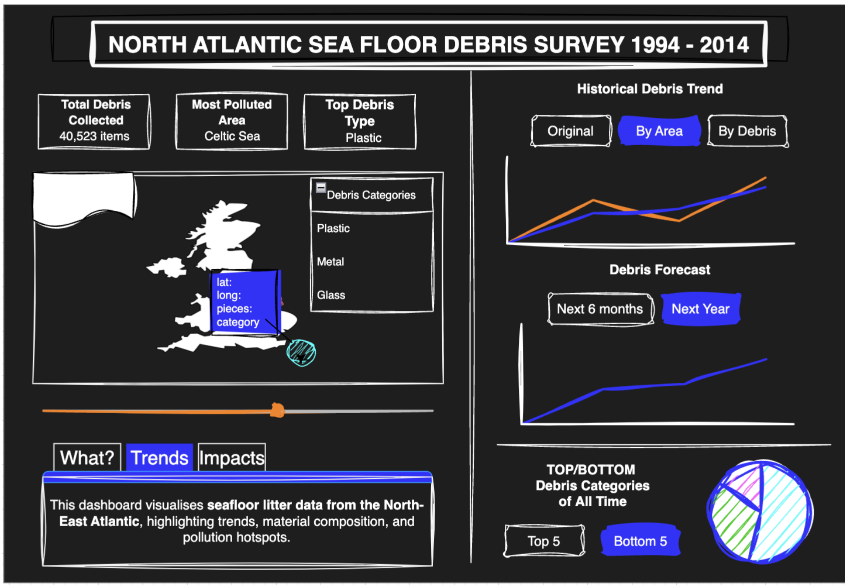
Enhancing Marine Litter Dashboard Design with AI Assistance

This report details the development of my marine litter dashboard, focusing on how AI tools, particularly Google Gemini, helped refine the design for clearer data communication. I ran into some challenges with my initial wireframe designs when trying to show data forecasts and trends, and mapping spatial data, therefore I used AI tools to help improve my visualisations and the overall usability of the dashboard.

Challenges in Initial Wireframes

The first version of my dashboard included a forecast feature meant to predict future marine litter trends based off of the historical data. But I realised the forecast models struggled because they didn't have enough data points between the values. The forecasts, similar to the ones exhibited in my notebook, were disconnected straight lines. The projections were too simplistic and didn't show the complexities of fluctuations in marine litter over time. I decided that an inaccurate forecast would be misleading and instead I could use visualisations related to historical data, focusing on the trends.



Beyond the forecasting flaw, the dashboard's layout needed some work too. At first, I thought animation on the map would be an interesting way to show the distribution of litter across the different areas, but then I realised it risked making the map too cluttered and a bit tricky for people to use. Plus, the user interface didn't have a clear structure, which would have made exploring the data difficult for my end users.

AI-Assisted Redesign Process

To address these challenges, I uploaded a photograph of my wireframe to Google Gemini for analysis. The AI provided valuable suggestions on improving the user interface and data visualisations to better fit stakeholder needs.

Refining Data Visualisations

One of the key takeaways from Gemini was the importance of showing changes in litter accumulation over time. Instead of a static trend line, the AI recommended animated line charts to illustrate gradual fluctuations in marine debris levels. This approach provided a more dynamic and accurate representation of trends and it would allow me to include an aspect of animation in the dashboard. For the spatial map, Gemini suggested shifting the focus from types of marine litter to a concentration based visualisation. Rather than displaying movement, the map would now highlight total litter accumulation across different locations, making it easier for my end users to compare litter density across the key areas and identify hotspot zones.

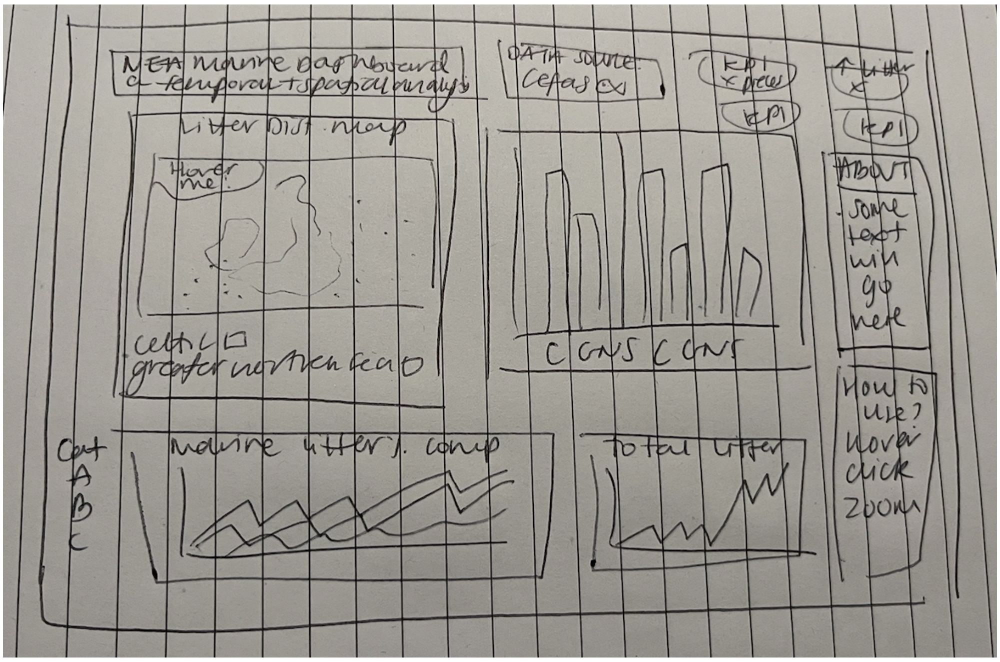
User Experience Improvements

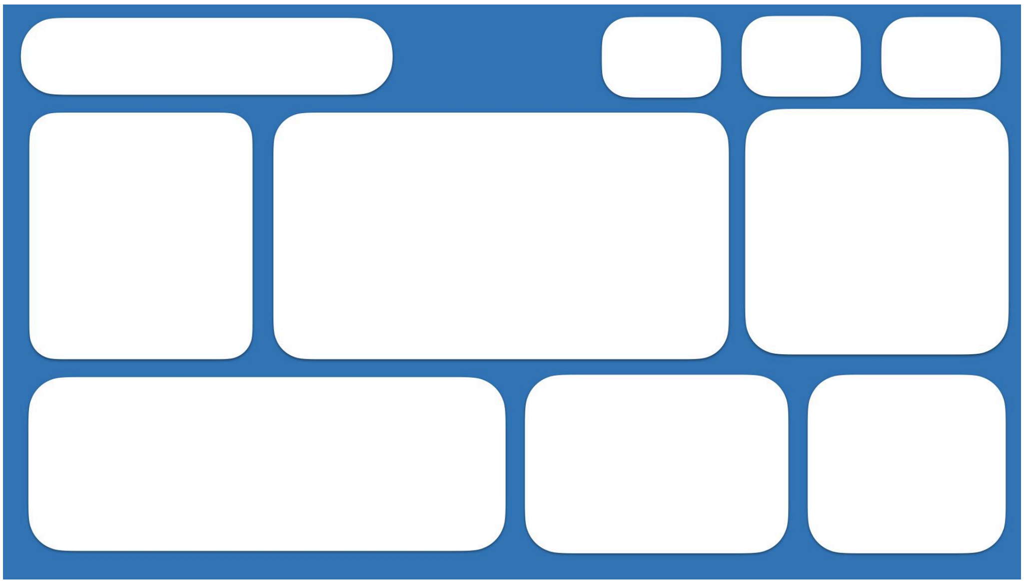
Gemini also provided insights into best practices for dashboard usability, recommending:

- Clearer labelling to ensure data points are easily understood.
- Boxed-off graph sections to visually separate different insights.
- Intuitive navigation to allow users to explore the data with minimal effort.
- Interactive elements such as tooltips and filters to enhance engagement.

From Sketches to Tableau

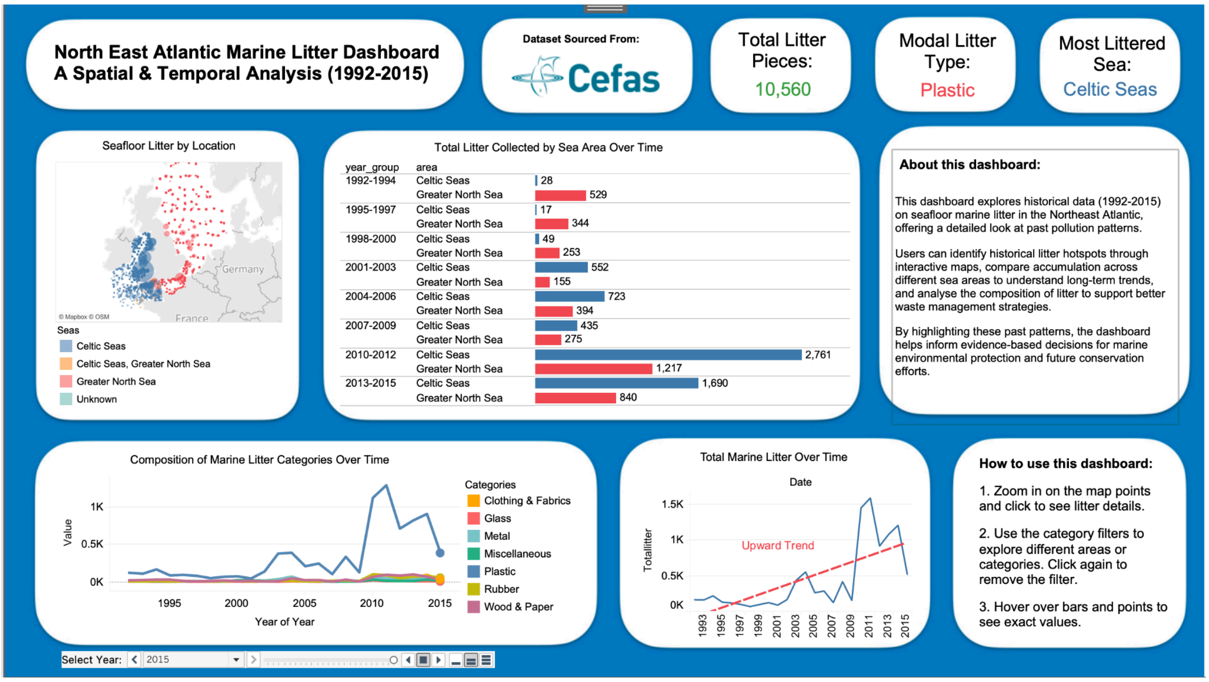
I found it easier to get my ideas onto paper to begin with, so I used pen and paper to sketch potential layouts for the map, charts, and data displays. This helped me quickly experiment with different configurations without committing and having to redo the dashboard design. Once I had a clear structure in mind, I used Keynote to create wireframe boxes, defining the placement of each element.





Using the Keynote wireframe as a guide, I then built the redesigned dashboard in Tableau, ensuring that the new layout and visualisations aligned with the AI-driven recommendations.

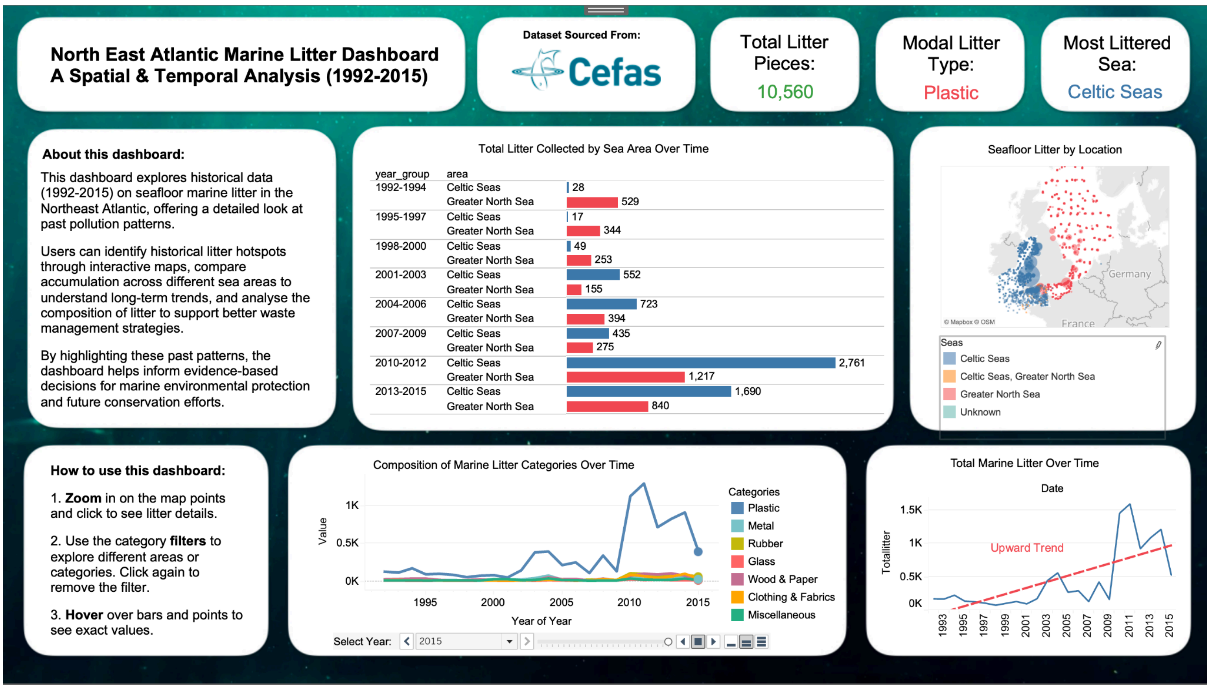
This version provides a clearer, more interactive experience for stakeholders, effectively communicating marine litter trends while avoiding misleading representations.



Following the AI-assisted redesign process, I focused on refining the dashboard's layout and visual aesthetics. I consulted with my cousin, a UX designer, who provided valuable insights on optimising the layout for better usability. She suggested using an F-style layout design, which arranges elements to match how people naturally read content from left to right, improving information flow and user engagement. Interestingly, this specific layout recommendation wasn't something the AI picked up, showing how human expertise and different perspectives can enhance AI-generated suggestions.

I initially considered a dark mode design but found the aesthetics didn't quite fit.

Instead, I chose a darker sea background, sourced from Pexels (https://www.pexels.com/@blaque-x-264516/). This sea background subtly enhances the dashboard's marine theme while maintaining readability.



Analysis of the Final Dashboard Design

The final dashboard effectively presents data in a clear and user-friendly manner.

Key components:

- Header and Key Information
  - The title clearly states the dashboard’s purpose and scope.
  - "Dataset Sourced From: Cefas" establishes data credibility.
  - Key performance indicators (KPIs) such as "Total Litter Pieces: 10,560," "Modal Litter Type: Plastic," and "Most Littered Sea: Celtic Seas" provide immediate insights of key information.

- Seafloor Litter by Location (Map)
  - The map provides a spatial representation of litter distribution.
  - Colour-coded markers for the different areas of the surveys.
  - The size of the point is reflective of the total litter count in that specific area.

- Total Litter Collected by Sea Area Over Time (Table)
  - This side-by-side barchart visualisation offers detailed comparisons of litter collection across different sea areas and timeframes.
  - It allows the user to see the exact number of pieces of litter collected.

- Composition of Marine Litter Categories Over Time (Line Graph)
  - This line graph effectively visualises the trends of different litter types over time.
  - Clearly labelled categories (Clothing & Fabrics, Glass, Metal, etc.) allow for easy interpretation.
  - The user can see which type of litter is increasing.

- Total Marine Litter Over Time (Line Graph)
  - This graph shows the overall trend of marine litter accumulation.
  - The "Upward Trend" annotation highlights the increasing problem.
  - The user can see that the amount of marine litter is increasing.

- About This Dashboard & How to Use It
  - These sections provide essential context and usage instructions, making the dashboard accessible to a broad audience.
  - The How to Use section gives three easy-to-follow steps.

- Select Year (Interactive Element)

    - This interactive element allows the user to change the year of the data being displayed.
-
- Visual Design
    - The dashboard uses a clean and organised layout, making it easy to navigate.
    - The use of white backgrounds for the individual graphs helps separate the information and make it easier to read.

Conclusion

Through this process, I learned several key lessons.

The first design isn’t necessarily the best design. The initial wireframes had usability issues, and refining the designs based on feedback led to better results. This highlights the importance of iterative design processes.

I also learnt that it is better to use a less sophisticated visualisation than use an inaccurate projection. It is better to focus on clear trend analysis when data constraints exist, especially when the focus is on conveying a trend and story to an audience. Similarly, remembering to keep things simple and not over cluttered. While animations would have been nice for the map, the amount of points and movement can be confusing for an audience.

Finally, I have learnt that AI can provide valuable recommendations, but critical thinking is still needed to assess and apply its suggestions effectively.

If I had more time, I would definitely iterate on the design further. That means I'd make more changes and improvements based on what I've learned. I'd also love to hold focus groups to see how people actually use the dashboard. It would be really helpful to watch how different people interact with it and get their feedback. That way, I could make sure it's as intuitive and useful as possible for everyone.